

Increased prevalence of constipation in pre-school children is attributable to under-consumption of plant foods: A community-based study.

AIM: To evaluate consumption of foods rich in dietary fibre and its relation to the prevalence of constipation in pre-school children. **METHODS:** In total, 368 children aged 3-5 years were randomly selected from kindergartens in Hong Kong. Constipation was confirmed by Rome-criteria. Children with normal bowel habits served as non-constipated controls. Consumption of vegetables, fruits, whole-grain cereals and fluid were determined using a 3-day food record. **RESULTS:** A total of 28.8% children were reported to have constipation. Median dietary fibre intake of constipated children was significantly lower than non-constipated children (3.4 g/d (inter-quartile range (IQR): 2.3-4.6 g/d) vs. 3.8 g/d (IQR: 2.7-4.9 g/d); $P = 0.044$) corresponding to 40% reference dietary fibre intake. Constipated children also had significantly lower intakes of vitamin C ($P = 0.041$), folate ($P = 0.043$) and magnesium ($P = 0.002$). Fruit intake and total plant foods intake were significantly lower in the constipated than non-constipated children: (61 g/d (IQR: 23.8-115 g/d) vs. 78 g/d (IQR: 41.7-144.6 g/d); $P = 0.047$) and (142.5 g/d (IQR: 73.7-214.7 g/d) vs. 161.1 g/d (IQR: 98.3-233.3 g/d); $P = 0.034$), respectively. Total fluid intake did not differ between groups but milk intake among the constipated children was marginally higher than the non-constipated children ($P = 0.055$). **CONCLUSION:** Insufficient dietary fibre intake is common in Hong Kong pre-school children. Constipated children had significantly lower intakes of dietary fibre and micronutrients including vitamin C, folate and magnesium than non-constipated counterparts which was attributable to under-consumption of plant foods. However, milk intake was marginally higher in the constipated children. More public education is necessary for parents to help develop healthy dietary habit and bowel habit in early life in order to prevent childhood constipation.